
Uploaded file(s)
G1 Decode magic bytes · EXIF orientation · HEIC → JPEG · PDF pages/text layer
G2 Quality Laplacian variance · brightness · glare · lighting · resolution · skew · cropping
G3 Preprocess Otsu mask → document quad → homography → deskew → shadow flattening → contrast
PDF text layer?
yes
no
Use embedded text confidence 1.0
G5 OCR Tesseract LSTM (PSM 4) words · boxes · confidences · auto-rotation search
G6 Layout rows · header/items/totals/footer · key-values · table header
G4 Classification bill vocabulary · money rows · dates vs ID/chat/letter/photo signals
not bill
bill / ambiguous
Polite rejection user may override
G7 Rule extraction labelled amounts · dates · merchant · ids · line items + evidence
G11/G12 VLM allowed AND (mode=always OR ambiguous OR weak required fields)?
yes
no
Vision-language model strict tool schema · untrusted-data prompt
Evidence-aware merge agree / disagree / model-only (verified in OCR?)
G8 Categorise lexicon rules + TF-IDF LR (registry) + VLM hint
G10 Validate arithmetic interpretations · dates · currency · items
G9 Confidence heuristic score · calibration table if registered · review states
Review-ready result schema billkit.extraction/1.0
G13 Human review → approved values → verified bill